

# An eleven-vertex tournament with no simultaneously optimal backedge ordering

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## Abstract

The clique number and chromatic number of a tournament's backedge graph can each be minimized over all vertex orders. We give an explicit tournament on eleven vertices for which no order attains both minima. Its minimum backedge clique number is 2, and its minimum backedge chromatic number is 3. Its unique order with a triangle-free backedge graph produces the Mycielski graph of the five-cycle, which has chromatic number 4. This answers negatively the simultaneous-ordering question of Aboulker, Aubian, Charbit, and Lopes, restated by Aubian and Coulomb. The uniqueness assertion is an exact finite computational proof: all  $11! = 39\,916\,800$  orders are tested, with a separate complete prefix enumeration and a direct backedge-graph recomputation. We give the input, completeness arguments, and reproduction instructions. No minimality or priority claim is made.

## 1 The question and the tournament

A *tournament* is an orientation of a complete graph. For a total order  $\prec$  on its vertices, the *backedge graph*  $T^\prec$  has the same vertices and has an undirected edge  $uv$  exactly when  $u \prec v$  and the tournament arc is  $v \rightarrow u$ . Write

$$\vec{\omega}(T) = \min_{\prec} \omega(T^\prec), \quad \vec{\chi}(T) = \min_{\prec} \chi(T^\prec), \quad (1)$$

where  $\omega$  and  $\chi$  are the ordinary graph clique and chromatic numbers. The second parameter is the dichromatic number: an independent set in a backedge graph is a transitive subtournament, and conversely concatenating topological orders of transitive parts gives an order whose backedge graph is colored by those parts.

Aboulker, Aubian, Charbit, and Lopes [1, Question 3.5] ask whether every tournament admits an order simultaneously attaining both minima in (1). Aubian and Coulomb restate this as Problem 7.11 of [2]. The contribution here is an explicit negative instance together with a complete computation excluding every simultaneous order. For adjacent context on vertex orders in graph theory, Zeng [3] studies Ramsey degrees for ordered split graphs. In that setting embeddings preserve the order of a fixed graph; here changing the order changes the backedge graph itself.

Let  $T$  have vertex set  $\{1, \dots, 11\}$ . For  $i < j$ , direct the pair as  $j \rightarrow i$  if  $\{i, j\}$  belongs to the following set, and as  $i \rightarrow j$  otherwise:

$$\begin{aligned} E = \{ & \{1, 5\}, \{1, 7\}, \{1, 8\}, \{1, 10\}, \{2, 4\}, \{2, 5\}, \{2, 9\}, \{2, 10\}, \\ & \{3, 7\}, \{3, 10\}, \{3, 11\}, \{4, 6\}, \{4, 7\}, \{4, 8\}, \\ & \{5, 11\}, \{6, 10\}, \{6, 11\}, \{7, 9\}, \{8, 11\}, \{9, 11\} \}. \end{aligned} \quad (2)$$

Thus every pair has exactly one direction. Put  $\prec_0 = (1, 2, \dots, 11)$  and  $G = T^{\prec_0}$ , so that  $E(G) = E$ .

**Theorem 1** (Computational disproof). *For the tournament  $T$  defined by (2),*

$$\vec{\omega}(T) = 2, \quad \vec{\chi}(T) = 3.$$

*The only order with triangle-free backedge graph is  $\prec_0$ , and  $\chi(T^{\prec_0}) = 4$ . Consequently,*

$$\min\{\chi(T^{\prec}) : \omega(T^{\prec}) = \vec{\omega}(T)\} = 4,$$

*and no order simultaneously attains the two minima in (1).*

## 2 Proof

**Lemma 2.** *The graph  $G$  is triangle-free and has chromatic number 4.*

*Proof.* Relabel its vertices by

$$(v_0, v_1, v_2, v_3, v_4) = (1, 10, 2, 4, 7), \quad (u_0, u_1, u_2, u_3, u_4) = (3, 5, 6, 9, 8), \quad w = 11.$$

Inspection of (2) gives exactly the following edges, with indices modulo 5: the  $v_i$  form a five-cycle; each  $u_i$  is adjacent to  $v_{i-1}$ ,  $v_{i+1}$ , and  $w$ ; there are no other edges. This is the Mycielski construction on  $C_5$ , also called the Grötzsch graph. We use the explicit description, so the identification requires no external graph catalogue.

The  $v$ -cycle contains no triangle. The two  $v$ -neighbors of each  $u_i$  are nonadjacent, and the neighbors of  $w$  form the independent set  $\{u_0, \dots, u_4\}$ . These observations exclude every possible triangle. To obtain a four-coloring, three-color the  $v$ -cycle, give  $u_i$  the color of  $v_i$ , and give  $w$  a fourth color.

Suppose that a three-coloring exists, and call the color of  $w$  color 3. Every  $u_i$  then has color 1 or 2. On the  $v$ -cycle replace the color of each  $v_i$  of color 3 by the color of  $u_i$ . On a cycle edge at most one endpoint is recolored. If it is  $v_i$ , the edge from  $u_i$  to the other endpoint ensures that the replacement color is different from that endpoint's color. We have produced a proper two-coloring of an odd cycle, a contradiction.  $\square$

**Lemma 3** (Exact enumeration). *The natural order  $\prec_0$  is the unique order of  $T$  with a triangle-free backedge graph.*

*Computational proof.* The complete permutation enumeration described in Section 3 tests all 39 916 800 orders. Exactly 39 916 799 have a backedge triangle, and its sole surviving order is  $(1, 2, \dots, 11)$ . A separately implemented complete prefix enumeration gives the same unique order. An additional implementation constructing each actual backedge graph reproduces the full-permutation count. The exact input, rejection criterion, coverage argument, and recorded counts are specified below. This lemma depends on these finite computations; it is not asserted as a computation-free structural result.  $\square$

*Proof of Theorem 1.* By Lemma 2,  $\omega(G) = 2$ , since  $G$  is triangle-free and has edges. Hence  $\vec{\omega}(T) \leq 2$ . The arcs  $1 \rightarrow 2 \rightarrow 7 \rightarrow 1$  form a directed triangle. An order with backedge clique number at most 1 would have no backedges, making  $T$  transitive. Therefore  $\vec{\omega}(T) = 2$ .

The following three lists partition the vertices:

$$(7, 10, 1, 11, 3), \quad (4, 5, 9, 2), \quad (6, 8). \quad (3)$$

All arcs within each list go forward, as follows directly from (2). Their concatenation is therefore an order whose backedge graph is properly colored by the three list labels. This proves  $\vec{\chi}(T) \leq 3$ . If

$\bar{\chi}(T) \leq 2$ , an order attaining this bound would have a bipartite, hence triangle-free, backedge graph. By Lemma 3, it would be  $\prec_0$ , contradicting  $\chi(G) = 4$ . Thus  $\bar{\chi}(T) = 3$ .

Since  $\bar{\omega}(T) = 2$ , every clique-optimal order has a triangle-free backedge graph. Lemma 3 makes  $\prec_0$  its only possibility, and Lemma 2 gives chromatic number 4 for that order. This proves all assertions.  $\square$

### 3 Exact computation and reproducibility

For the implementations only, subtract 1 from every vertex label. List the pairs  $(i, j)$  with  $0 \leq i < j < 11$  in lexicographic order, starting the bit index at 0. Bit 1 means  $j \rightarrow i$ , and bit 0 means  $i \rightarrow j$ . In this convention the tournament is encoded by

$$M = 11356307831593320.$$

The supplementary input also gives the one-based edge list in (2), and the Python verifier checks that the two encodings agree.

**Enumeration of all permutations.** Let  $A(x, y)$  mean that  $x \rightarrow y$  is an arc. Form the ordered triples

$$\mathcal{C} = \{(x, y, z) : A(x, y), A(y, z), A(x, z)\}.$$

For an order  $p$ , let  $r_p(x)$  be the position of  $x$ . Reject  $p$  exactly when some  $(x, y, z) \in \mathcal{C}$  satisfies

$$r_p(z) < r_p(y) < r_p(x). \quad (4)$$

Indeed, these three reversed arcs give a backedge triangle. Conversely, reading any backedge triangle from last vertex to first produces such a transitive triple. Thus (4) is an exact test, including both implications.

The C++ program starts at the ascending permutation and applies `std::next_permutation` until it returns false. Every permutation is visited once; there is no symmetry reduction, prefix pruning, or search cutoff. For each rejection the program increments the counter for its first violated triple. The sum of those counters is 39 916 799, and the only surviving permutation is  $(0, 1, \dots, 10)$ . All counters use 64-bit integers.

**Enumeration of all valid prefixes.** The Python program begins with the empty prefix and tries every unused vertex  $v$  as the next vertex. For an already triangle-free prefix, appending  $v$  creates a triangle exactly when the prefix contains  $u$  before  $w$  with

$$A(v, u), \quad A(v, w), \quad A(w, u).$$

These are precisely the three backward arcs involving the new vertex; a triangle not using  $v$  would already be in the prefix. Rejecting exactly these extensions visits all and only triangle-free prefixes, by induction on their length. The completed enumeration gives:

|                |        |        |        |        |       |        |
|----------------|--------|--------|--------|--------|-------|--------|
| Prefix length  | 0      | 1      | 2      | 3      | 4     | 5      |
| Valid prefixes | 1      | 11     | 110    | 874    | 5 166 | 21 258 |
| Prefix length  | 6      | 7      | 8      | 9      | 10    | 11     |
| Valid prefixes | 57 046 | 92 377 | 80 429 | 28 982 | 1 413 | 1      |

The sole full prefix is again  $(0, 1, \dots, 10)$ . The program also checks the explicit graph identification and the transitive lists, and obtains  $\chi(G) = 4$  and  $\bar{\chi}(T) = 3$  by independent subset-partition dynamic programs. Those dynamic programs are additional checks; the elementary arguments above establish the deductions from the uniqueness lemma.

**Direct backedge-graph check.** A third C++ implementation starts from the twenty explicit pairs rather than the integer mask. For every permutation it constructs the actual undirected backedge adjacency. It tests for a triangle by checking whether any edge has endpoints with a common neighbor. This avoids the transitive-triple and prefix rejection implementations. It again checks 39 916 800 orders and finds exactly the natural order triangle-free. It shares the C++ permutation-library routine with the first program; the Python program uses a different enumeration route.

**Reproduction.** The accompanying `supplement/` directory contains the input, three verifier sources, saved computational outputs, and a Python wrapper. From that directory run

```
python3 reproduce.py
```

The wrapper needs Python 3 and GCC or Clang with C++17 support. It compiles and runs both C++ verifiers, runs the Python verifier on copies of the supplied input, and compares the mathematical outputs with the saved records. Generated files stay in `reproduction/`. All mathematical tests use integers and Boolean adjacency. No external graph library, graph catalogue, SAT solver, or random sampling is used in these verifications.

## 4 Scope and assistance

The result concerns the explicitly specified tournament and disproves the universal simultaneous-ordering assertion. It does not classify other tournaments or establish a smallest counterexample. The literature comparison is limited to the inspected versions of the two question sources [1, 2]; bibliographic priority is not asserted.

The proof of Lemma 3 relies on ordinary compiler and runtime behavior and completed exact finite enumerations. No proof-assistant formalization is claimed. OpenAI Codex assisted with counterexample search, program development, mathematical exposition, and manuscript preparation. A separate model review reran the two supplied verifiers and wrote the direct backedge-graph verifier. That review is an additional computational and argument check, not external peer review or a formal proof certificate.

## References

- [1] P. Aboulker, G. Aubian, P. Charbit, and R. Lopes, *Clique number of tournaments*, arXiv:2310.04265v2 (2026; original preprint 2023). <https://arxiv.org/abs/2310.04265v2>.
- [2] G. Aubian and S. Coulomb, *Clique Number of Tournaments II*, arXiv:2609.07481v1 (2026). <https://arxiv.org/abs/2609.07481v1>.
- [3] Zijian Zeng, *The Small Ramsey Degree of an Edge in Ordered Split Graphs Is Three*, Preprints (2026). <https://doi.org/10.20944/preprints202608.2162.v1>.